Liem Fairbairn

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EDUCATION

Imperial College London — MEng Aeronautical Engineering (4-year Integrated Master's) 2025 - 2029
Year 1 Final Grade: 3.6/4.0 GPA (US Equivalent, UK Predicted 1st Class)

Alexandra Park School 2018 - 2025
A Levels: Mathematics (A*), Physics (A*), Classical Civilisation (A*), Further Mathematics (A)

TECHNICAL SKILLS

CAD & Design (CATIA Dassault 3DExperience, Fusion 360), Drafting (BS8888, ISO Based), Abaqus (FEA), Programming (MATLAB, Python, C++), KiCAD (PCB Design), STAR-CCM+ (CFD), Experimental Testing & Data Analysis

EXPERIENCE

Recovery Systems Engineer - Imperial College Space Society 2026 - Present

- Recovery lead for student-built rocket aiming for 2000 ft on a 29mm G motor with June 2027 flight.
- Mechanical design of recovery ejection system, working with airframe sub team for subsystem integration.
- Sizing Parachute and packing volume for sub 15 m/s landing velocity, <1km drift in 15 m/s crosswind and deployment at apogee, further developing recovery bay layout as well as attachment points.

Structural Engineer - Imperial College Space Society 2026 - Present

- Designing and developing 2-axis flywheel control demonstrator, flywheel and motor sizing, build and testing in progress (target Feb 2027)
- Sizing flywheels against motor-torque/momentum limits, developing 3D print-ready 10x10x10 cm envelope model in Fusion 360

ENGINEERING PROJECTS

HAWT Design and Manufacturing 2026

- Designed, manufactured and wind tunnel tested two turbine blades in a team of 6, extracting 13.9 W in a 10 m/s wind tunnel at 1600 rpm, manufactured with carbon fibre, 3D printed PLA, plywood, and foam.
- Analysed carbon-fibre spar structurally and produced CATIA drawings of multi-material blade sections.
- Evaluated factors of material, cost, time, manufacturing, aerodynamic, and structural properties to develop multi-material (PLA, Carbon fibre, Plywood, and foam) final design.
- Developed MATLAB Blade Element Momentum solver with aerodynamic correction models to predict and validate performance, down selecting aerofoil configurations.

NASA TechLeap Robotically Manipulated Payload Challenge - Independent Submission 2026

- Developed an engineering proposal for an in-space electric propulsion propellant (xenon) transfer payload, independently authoring a full thermodynamics/structural systems requirement addressing NASA propulsion shortfall 23.07.
- Produced CAD concept assembly (Dassault 3DEXPERIENCE), cost/budget narrative, engineering drawings, and video pitch as sole engineer and systems owner on the submission.
- Modelled supercritical blowdown dynamics using a real-gas Redlich-Kwong equation of state in MATLAB for verification and verified dual pressure vessel margins (6061-T6 aluminium, 42 MPa worst-case design pressure), for Safety Factor of 3 and 0.71 kg of Xenon Transferred.

Custom Drone Flight Controller Design (Ongoing) 2026 - Present

- Designing custom STM32F411-based flight controller integrating an ICM-42670-P IMU, MLX90394 magnetometer and BMP390 barometer for inertial, magnetic, and altitude sensing.
- Designed electronics architecture and KiCAD Schematic/PCB including power regulation and decoupling, SPI and I2C communications and sensor interrupt interfaces.

LABORATORY WORK

Structural and Material Analysis 2025 - 2026

- Built and mesh-converged 1D and 3D Abaqus FE models of statically indeterminate steel beam under 40N load. Validated against Euler-Bernoulli theory and physical strain gauge testing. Achieved convergence to 4mm at 28 elements
- Tensile tested aluminium and polymer specimens to fracture, identifying Young's Modulus and deformation behaviour.

Aerodynamics and Aircraft Design 2025 - 2026

- Derived stability and control derivatives (neutral point, downwash gradient $d\varepsilon/d\alpha = 0.96$, tailplane lift curve slope = 1.5/radian) of aircraft model via wind tunnel and flight simulation testing across 3 tailplane settings confirming a stable pitch break at stall.
- Built CFD model (STAR-CCM+) of laminar pipe flow to assess validation against experimental and Hagen-Poiseuille theory, achieving 3.71% agreement in pressure-coefficient gradient.